Machinery Safety Specialist (CE)

TAB3: Risk assessment
Module 1: Risk assessment process

Tab3-1/2 20220901

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Risk assessment
According to machinery directive -1

Recital 23:

- The manufacturer or his authorized representative should also ensure that a risk assessment is carried out for the machinery which he wishes to place on the market.
- For this purpose, he should determine which are the essential health and safety requirements applicable to his machinery and in respect of which he must take measures.

Risk assessment
According to machinery directive -1

Annex I, General principles

The manufacturer of machinery or his authorized representative must ensure that a risk assessment is carried out in order to determine the health and safety requirements which apply to the machinery. The machinery must then be designed and constructed taking into account the results of the risk assessment.

By the iterative process of risk assessment and risk reduction referred to above, the manufacturer or his authorized representative shall:

See step-by-step plan for risk assessment in accordance with EN-ISO 12100 as well as risk reduction (Tab 4).

Risk assessment
(European) standards

Standards for risk assessment

- *EN-ISO 12100: Safety of machinery - General principles for design - Risk assessment and risk reduction*

  Risk assessment and reduction strategy

- *NPR-ISO/TR 14121-2: Safety of machinery - Risk assessment - Part 2: Practical guidance and examples of methods:*
  - *Risk matrix*
  - *Risk graph*
  - *Numerical scoring*
  - *Hybrid tool*

Terms and definitions (1) (EN-ISO 12100)

- **Harm:** Physical injury or damage to health.
- **Hazard:** Potential source of harm.
- **Hazardous situation:** Circumstance in which a person is exposed to at least one hazard. The exposure can result in harm immediately or over a period of time.
- **Risk:** Combination of the probability of occurrence of harm and the severity of that harm.
- **Residual risk:** Risk remaining after protective measures have been implemented.

Terms and definitions (2) (EN-ISO 12100)

- **Risk assessment:** Overall process comprising a risk analysis and a risk evaluation.
- **Risk analysis:** Combination of the specification of the limits of the machine, hazard identification and risk estimation.
- **Risk evaluation:** Judgment, based on risk analysis, of whether the risk reduction objectives have been achieved.

Risk assessment machines

- Risk = probability x severity
- Definition risk assessment:

"A comprehensive estimate of the probability of injury or damage to health and the severity of the possible injury or damage to health in order to select the appropriate safety measures".

Risk assessment process

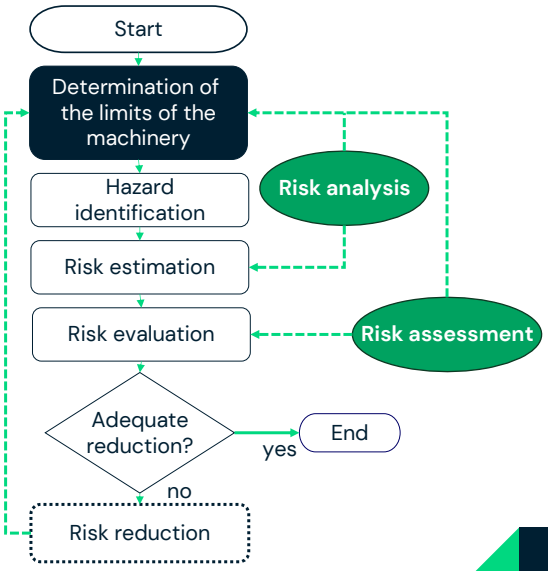
EN ISO 12100

Step 1:
Determine the limits of the machinery

Step 2:
Identify the hazards

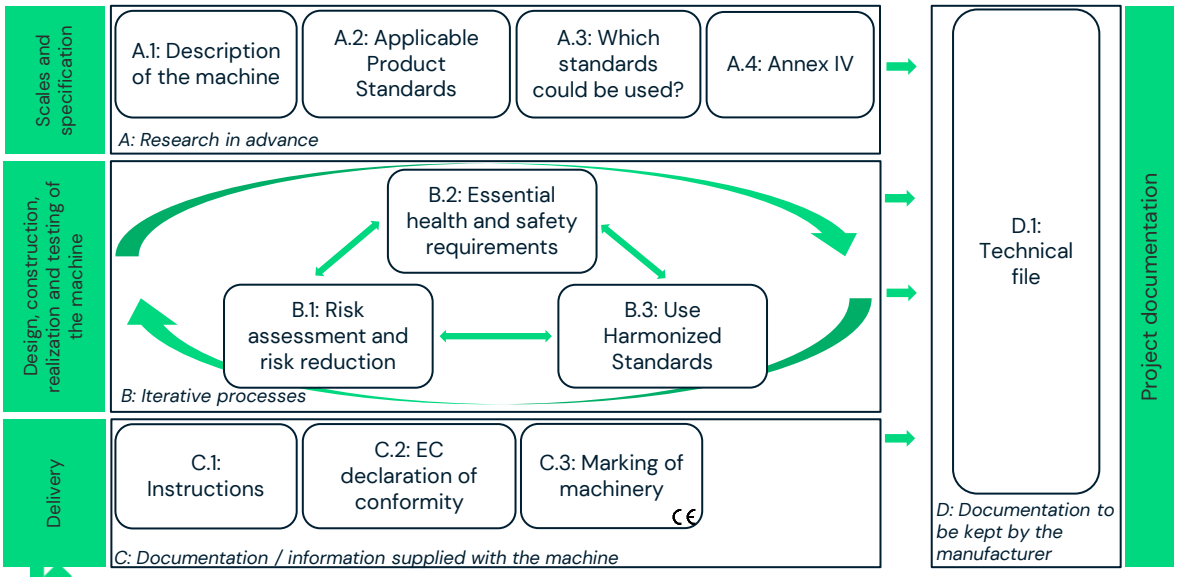
Step 3:
Estimate the risk

Step 4:
Evaluate the risk



Risk assessment process

EN ISO 12100

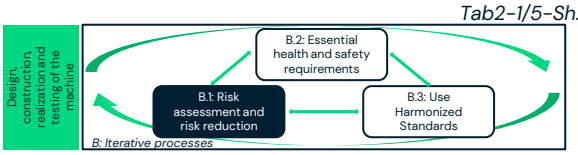


Tab2-1/5-Sh.9

Tab2-1/5-Sh.10

Risk assessment process

EN ISO 12100 – Determination of limits



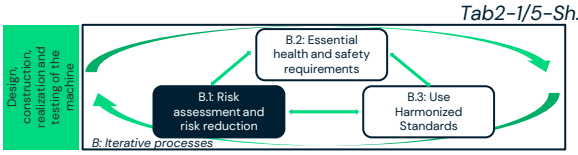
Use limits (simplified)

- What is the intended and reasonably foreseeable misuse of the machine?
- Modes of operation?
- How to intervene in the event of a malfunction?
- How and where is the machine used?
- Who uses the machine?
- Can the machine be used by people with limited physical abilities?
- Is training and/or experience required to operate the machine?
- Are others exposed to hazards besides the operator(s)?

Tab2-1/5-Sh.11

Risk assessment process

EN ISO 12100 – Determination of limits



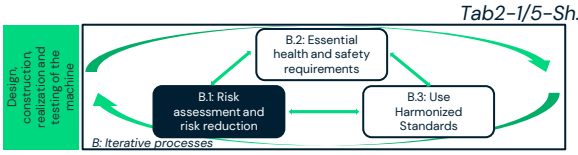
Space limits (simplified)

- What is the machine's range of movement?
- How much space does the machine require?
- How much space do the people working with the machine need?
- Where should operators or technicians perform activities?
- How is the machine operated (where are controls located)?
- How is the machine supplied with energy?
- Layout / sketch

Tab2-1/5-Sh.12

Risk assessment process

EN ISO 12100 – Determination of limits



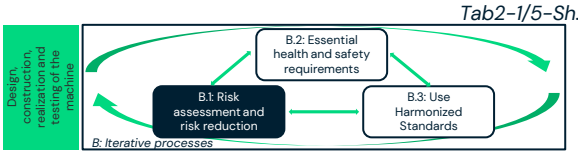
Time limits (simplified)

- What is the lifespan of the machine?
- Which wear-sensitive parts are regularly replaced?
- What are the expected service intervals?
- What is the expected intensity of use of the machine?
- Make realistic assumptions when something is not clear up front.
- These principles can be changed and updated during the design process of the machine.

Tab2-1/5-Sh.13

Risk assessment process

EN ISO 12100 – Determination of limits



Other limits (simplified)

- What are the environmental conditions?
- How does the machine get cleaned and how often is this necessary?
- What are the properties of the materials that are processed in the machine?

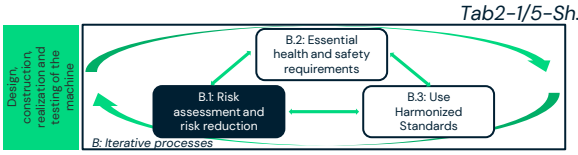
<u>How to obtain information as input for the determination of limits:</u>

- Involve service technicians involved in commissioning
- Involve operators (if possible/relevant)
- Accident history

Tab2-1/5-Sh.14

Risk assessment process

EN ISO 12100 – Determination of limits



Example: determination of limits for a fictional machine 'robot cell'

Brainstorming together about a number of possible limits of the the machine.

- What kind of users can be expected?
  -
  -
- At what locations can users be expected?
  -
  -
- Machine usage frequency?
  -
  -
- Environmental conditions?
  -
  -

Exercise 2

- Describe three relevant machine limits for the case you choose for your homework.
- Plenary discussion.

Risk assessment process

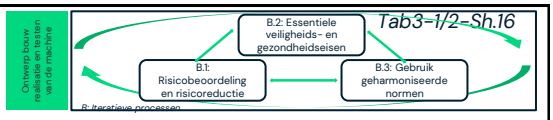
EN ISO 12100 – Hazard identification

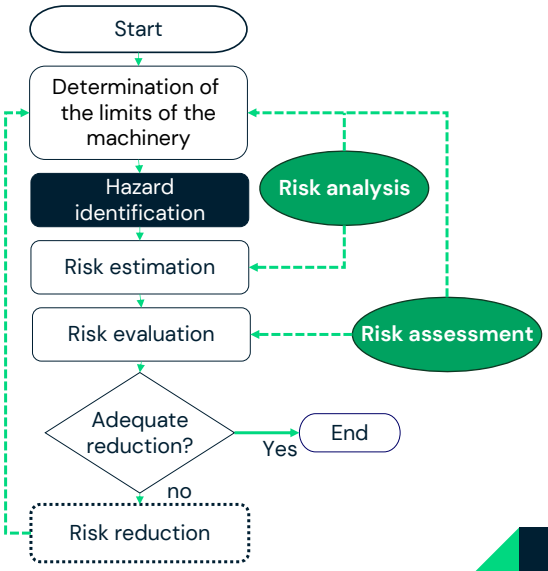
Step 1:
Determine the limits of the machinery

Step 2:
Identify the hazards

Step 3:
Estimate the risk

Step 4:
Evaluate the risk

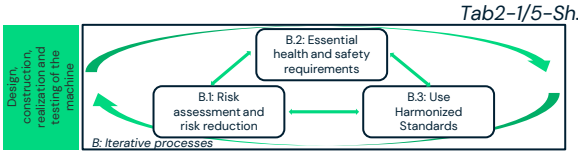




Tab2-1/5-Sh.17

Risk assessment process

EN ISO 12100 – Hazard identification



B.1: Risk assessment and risk reduction

Hazard Identification

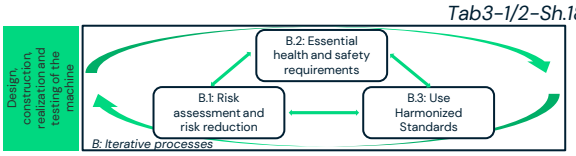
'The systematic identification of reasonably foreseeable hazards, hazardous situations and/or hazardous events during all phases of the machine life cycle'.

- Use a tailored template to record the risk assessment (in particular the identified hazards, the estimated risks and the necessary measures).
- Encode or number all risks so that they can always be referred to.
- If necessary, use Table B1 from EN ISO 12100 as a guideline for possible types of danger.
- In addition, use the applicable standards to determine whether certain hazards are applicable or not.

Tab3-1/2-Sh.18

Risk assessment process

EN ISO 12100 – Hazard identification



B.1: Risk assessment and risk reduction

Hazard identification (EN ISO 12100 Tabel B1)

1. Mechanical
2. Electrical
3. Thermal
4. Noise
5. Vibration
6. Radiation
7. Material/substance
8. Ergonomic
9. Environmental
10. Combinations of hazards

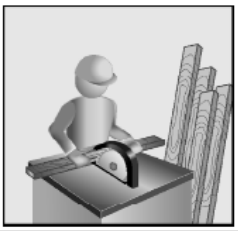
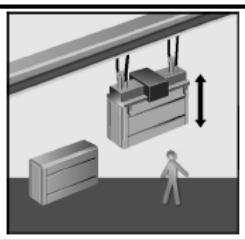
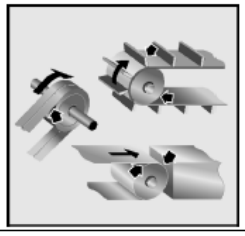
No	Type or group	Examples of hazards	
		Origin[a]	Potential consequences[b]
1	Mechanical hazards	— acceleration, deceleration; — angular parts; — approach of a moving element to a fixed part; — cutting parts; — elastic elements; — falling objects; — gravity; — height from the ground; — high pressure; — instability; — kinetic energy; — machinery mobility; — moving elements; — rotating elements; — rough, slippery surface; — sharp edges; — stored energy; — vacuum.	— being run over; — being thrown; — crushing; — cutting or severing; — drawing-in or trapping; — entanglement; — friction or abrasion; — impact; — injection; — shearing; — slipping, tripping and falling; — stabbing or puncture; — suffocation.

Tab3-1/2-Sh.19

Risk assessment process

Hazard identification

Table B2 EN-ISO 12100: 2010 examples

Hazard		Hazard	
	**Origin** cutting parts **Potential consequences** – cutting – severing		**Origin** falling objects **Potential consequences** – crushing – impact
	Origin moving elements **Potential consequences** – crushing – impact – shearing		**Origin** moving elements (three examples) **Potential consequences** – drawing-in – friction, abrasion – impact

Tab3-1/2-Sh.20

Risk assessment process

Hazard identification

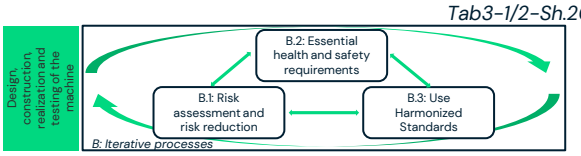


Table B3 EN-ISO 12100 : 2010 Examples / Tasks

Operation	— Clamping/fastening the workpiece — Control/inspection — Driving the machine — Feeding, filling, loading of raw material — Manual loading/unloading — Minor adjustments and setting of functional parameters of the machine (for example, speed, pressure, force, travel limits) — Minor interventions during operation (for example, removing waste material, eliminating jams, local cleaning) — Operating manual controls — Restarting the machine after stopping/interruption — Supervision — Verification of the final product
Cleaning **Maintenance**	— Adjustments — Cleaning, disinfection — Dismantling/removal of parts, components, devices of the machine — Housekeeping

Tab3-1/2-Sh.21

Risk assessment process

Hazard identification

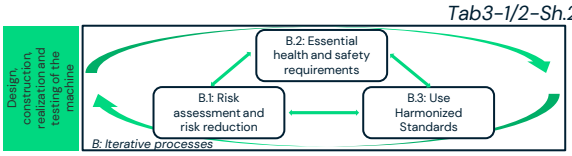
Table B4 EN-ISO 12100: 2010 hazardous events

Origin related to...	Hazardous event	Subclause of this International Standard
Shape and/or superficial finishing of accessible parts of the machine	— Contact with rough surfaces — Contact with sharp edges and corners, protruding parts	6.2.2.1
Moving parts of the machine	— Contact with moving parts — Contact with rotating open ends	6.2.2, 6.2.14, 6.2.15 6.3.1 to 6.3.3 6.3.5.2 to 6.3.5.4 6.4.3 to 6.4.5
Control system	— Dropping or ejection of a moving part of the machine or of a workpiece clamped by the machine — Failure to stop moving parts — Machine action resulting from inhibition (defeating or failure) of protective devices — Uncontrolled movements (including speed change) — Unintended/unexpected start-up — Other hazardous events due to failure(s) or poor design of the control system	6.2.5 6.2.11 to 6.2.13 6.3.5.2 to 6.3.5.4 6.4.3 to 6.4.5

Tab3-1/2-Sh.22

Risk assessment process

EN ISO 12100 - Hazard identification



Type-C standard?

- *Type-C standards contain lists of significant hazards*
- *Usually in chapter 4*
- *Very specific, sometimes with images*
- *See, for example, the standard for pallet wrappers (EN 415-6)*

NB Many type-C standards also include a list of safety measures (which are specific for that type of machine).

Tab3-1/2-Sh.23

Risk assessment process

EN ISO 12100 - Hazard identification

Recording in RA-template

Hazard identification (example template)

ID	Use phase	Operational modes	User / Exposed Persons	Task / Action	Scenario factors			
					Hazard	Hazard zone	Hazardous situation	Hazardous event
2								

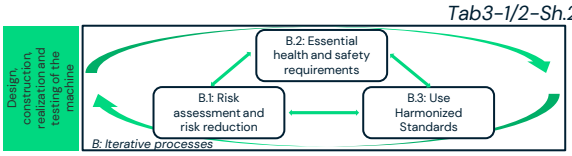
Important! Consistency with limits of the machinery

Tip: start with hazardous situation / event (think in scenario's).

Tab3-1/2-Sh.24

Risk assessment process

EN ISO 12100 - Hazard identification



Recording in RA-template

Example: hazard identification fictive machine 'robot cell)

ID	Use phase	Operational modes	User / Exposed Persons	Task / Action	Scenario factors			
					Hazard	Hazard zone	Harardous situation	Harardous event
1	Production	Automatic mode	Operator	Feeding product on convayor belt	Impact or crushing hazard	Input side robotcell	Operator wants to grab a lost object from the convayor belt and reaches through an opening in the guard.	Broken bones in the hand of he operator because he was hit by the robot.

Tab3-1/2-Sh.25

Exercise 3

- Record three relevant hazards for your use case. Use the example list from Annex B of the EN-ISO 12100 standard.
- Plenary discussion.

Tab3-1/2-Sh.26

Risk assessment process

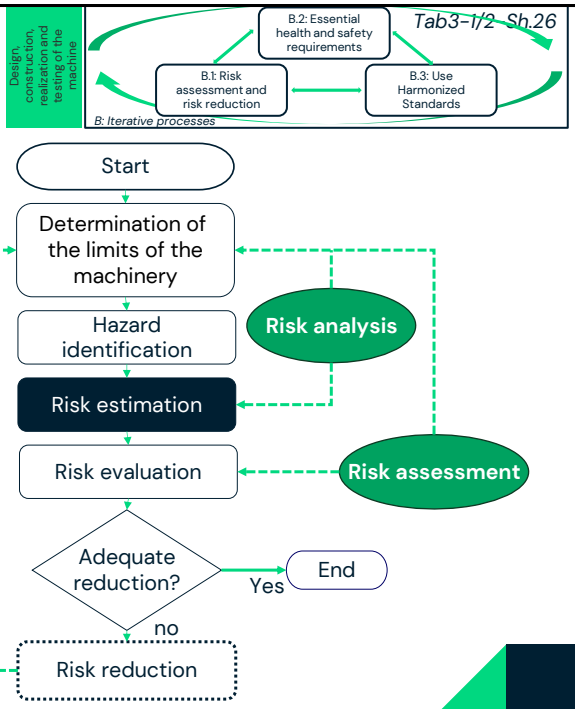
EN ISO 12100 – Risk estimation

Step 1:
Determine the limits of the machinery

Step 2:
Identify the hazards

Step 3:
Estimate the risk

Step 4:
Evaluate the risk



Tab3-1/2-Sh.27

Risk assessment process

EN ISO 12100 – Risk estimation

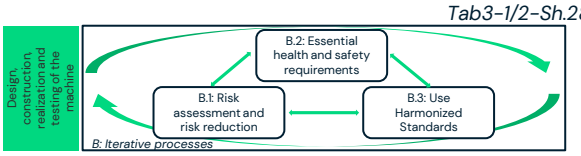
Risk Estimation by estimating the following elements:

- Severity of harm
- Probability of occurrence of that harm, which is a function of:
  - the exposure of person(s) to the hazard
  - the occurrence of a hazardous event
  - the technical and human possibilities to avoid or limit harm.

Tab3-1/2-Sh.28

Risk assessment process

EN ISO 12100 – Risk estimation



Severity of harm

Severity can be estimated by taking the following into account :

The severity of injuries or damage to health, for example:

- slight
- serious
- death

The extent of harm, for example, to:

- one person
- several persons

Tab3-1/2-Sh.29

Risk assessment process

EN ISO 12100 – Risk estimation

Exposure of persons to the hazard

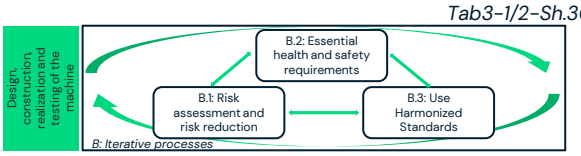
Factors to be taken into account when estimating the exposure are, amongst others:

- the need for access to the hazard zone (for normal operation, malfunctions, maintenance, repair, etc.),
- the nature of access (for example, manual feeding of materials)
- time spent in the hazard zone
- the number of persons requiring access
- the frequency of access

Tab3-1/2-Sh.30

Risk assessment process

EN ISO 12100 – Risk estimation



Occurrence of a hazardous event

Factors to be taken into account when estimating the occurrence of a hazardous event are, among others:

- reliability and other statistical data
- accident history
- historical data on health damage
- comparison of risks (similar machines, hazards, specifications, conditions of use, etc.)

Tab3-1/2-Sh.31

Risk assessment process

EN ISO 12100 – Risk estimation

Possibility of avoiding or limiting harm:

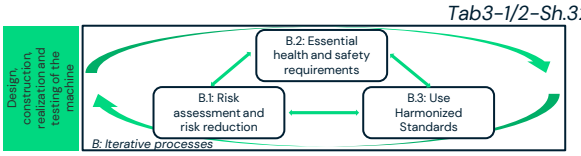
Factors to be taken into account when estimating the possibility of avoiding or limiting harm are, among others, the following:

- education of persons who can be exposed to the hazard(s)
- how quickly the hazardous situation could lead to harm
- any awareness of risk (information for use, direct observation, warning signs and indicating devices)
- the human ability to avoid or limit harm (reflexes, agility)
- practical experience and knowledge (of (similar) machinery)

Tab3-1/2-Sh.32

Risk assessment process

EN ISO 12100 – Risk estimation



Aspects to be considered during risk estimation:

Persons exposed (all persons (operators and others) for whom exposure to the hazard is reasonably foreseeable

- Type, frequency and duration of exposure (e.g., needs for access during normal operation and maintenance)
- Relationship between exposure and effects (Accident data can assist)
- Human factors (e.g., stress, ergonomic aspects)
- Suitability of protective measures
- Possibility of defeating or circumventing protective measures (see EN-ISO 14119, tab 4)
- Ability to maintain protective measures
- Information for use

Tab3-1/2-Sh.33

Risk assessment process

EN ISO 12100 – Risk estimation

Recording in RA-template

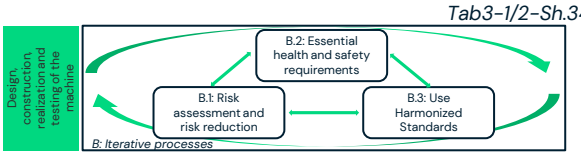
Risk estimation (example template)

Explanation risk assessment	S	F	W	A	Risk	Protective measures				S	F	W	A	Risk	Risk evaluation	Remark
Summary of the scenario factors (hazard analysis) with an explanation of the risk assessment.	S1 S2 S3	F1 F2	P1 P2 P3	A1 A2	1 - 14	STEP 1 Inherently safe design measures	STEP 2a Safeguarding and/or complementary protective measures	STEP 2b Identification safety functions	STEP 3 Information for use	S1 S2 S3	F1 F2	P1 P2 P3	A1 A2	1 - 14	(Residual) risk acceptable? Y/N (according to EN 12100 5.6.2)	remark / explanation

Tab3-1/2-Sh.34

Risk assessment process

EN ISO 12100 – Risk estimation



Recording in RA-template

Example: Risk estimation fictive machine 'robot cell'

Explanation of the risk assessment methods follows in module 3.2

ID	Haz. Nr.	Explanation risk assessment	S	F	W	A	Risk
	Harard from table B.1 van ISO 12100	Summary of the scenario factors (hazard analysis) with an explanation of the risk assessment.	S1 S2 S3	F1 F2	P1 P2 P3	A1 A2	1 - 14
1		S2: Broken bones in hands and/or fingers. F2: Constantly operational W2: Possible that an operator reaches through the guard in an unguarded moment. P2: Movement of robot is fast. Direction of movement not always known by the operator.	S2	F2	W2	P2	8

Risk assessment process

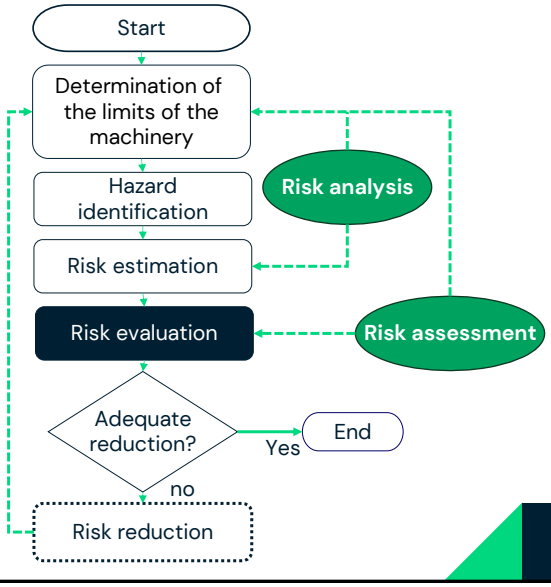
EN ISO 12100 – Risk estimation

Step 1:
Determine the limits of the machinery

Step 2:
Identify the hazards

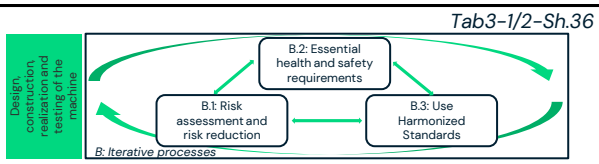
Step 3:
Estimate the risk

Step 4:
Evaluate the risk



Risk assessment process

EN ISO 12100 – Risk estimation



Risk evaluation and Risk reduction:

- Basic principle: Eliminate dangers as much as possible or reduce risks as much as possible.
- Manufacturer is responsible for determining whether a residual risk is acceptable or not.
- Using (harmonized) Standards is convenient since that is where the 'accepted' minimum safety level is laid down [B.3]
- Record in the risk assessment why this residual risk is acceptable; for each residual risk (otherwise further risk reduction is required).
- Define risk-reducing measures for each risk in accordance with the 'mandatory sequence' (see tab 4)

Make sure that measures do not increase the risk and that no new hazards are introduced by the measure.

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